Mapping Water Stress and Agricultural Adaptation Potential Using Multi-Source 16-Band Attention U-Net (Case of Taroudant, Morocco)

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***Abstract*—Water scarcity represents a major and growing constraint on agricultural production across arid and semi-arid regions, a challenge significantly intensified by global climate change and the resulting increase in drought frequency. This study addresses the critical need for high-resolution, actionable water stress mapping in the Taroudant region of Morocco, a known water stress hotspot within the Souss-Massa basin. We propose a novel, weakly supervised framework utilizing an Attention U-Net deep learning architecture. The model is trained on a unique 16-band multi-source remote sensing data stack, fusing optical, radar, soil moisture, climate, and socio-economic variables. Crucially, this framework produces detailed maps of agricultural water stress at a 3 km resolution, providing a regional overview essential for large-scale adaptation planning. A Smart Health Index (SHI), derived from satellite indicators, serves as the training label, effectively overcoming the logistical and financial barriers associated with extensive field data collection. The resulting maps classify stress into Severe, Moderate, and Healthy categories, offering critical insights into key drivers such as root-zone soil moisture and plant water content. This work significantly contributes to data-driven decision-making for regional water resource management, targeted irrigation scheduling, and the implementation of effective, drought resilient agricultural practices. While high-resolution inputs are processed at 3 km to capture field-level spatial variability, the resulting stress maps are interpreted at a coarser effective scale, consistent with the spatial support of root-zone soil moisture observations, enabling robust regional-scale analysis.**

***Index Terms*—Attention U-Net, Drought mapping, Google Earth Engine, Remote sensing, SMAP, Taroudant, Water stress.**

I. INTRODUCTION

Water scarcity represents one of the most pressing environmental challenges of the 21st century, particularly in arid and semi-arid regions where agricultural productivity is heavily dependent on water availability. Climate change has exacerbated this issue through altered precipitation patterns, increased temperatures and more frequent extreme events. The Mediterranean region, including Morocco, has been identified as a climate change hotspot, with significant reductions in water availability and increased drought frequency [1]–[3].

Remote sensing has emerged as a powerful tool for monitoring agricultural systems and water resources at multiple spatial and temporal scales. The integration of satellite-derived indices with advanced machine learning methods enables precise mapping of water stress conditions and supports data-driven decision-making for water resource management [4]–[7].

Taroudant province in southern Morocco exemplifies these challenges. The region is characterized by a semi-arid to arid climate, intensive irrigated citrus orchards and traditional olive groves, all under growing stress from groundwater depletion and rainfall variability [1], [2], [8]. This paper proposes an end-to-end framework that combines Google Earth Engine (GEE) data processing with an Attention U-Net model to produce field-scale (3 km) water stress and adaptation maps over Taroudant.

II. RELATED WORKS

Recent years have seen increasing interest in data-driven and deep learning approaches for agricultural drought and water stress monitoring using remote sensing data. Traditional indices such as the Normalized Difference Vegetation Index (NDVI), the Normalized Difference Water Index (NDWI) and the Soil Adjusted Vegetation Index (SAVI) are widely used as proxies for vegetation health, plant water content and soil-adjusted greenness [4]–[6], [9]. However, NDVI alone often fails to capture root-zone moisture conditions and may remain high even when plants are under hidden water stress [7], [10].

Multi-sensor data fusion has therefore gained importance, combining optical indices with thermal information, radar backscatter and coarse-resolution soil moisture products to improve drought characterization [6], [7], [11], [12]. In parallel, convolutional neural networks (CNNs) and U-Net variants have been increasingly adopted for segmentation problems in agriculture, including crop type mapping, yield estimation and disease detection, due to their ability to model spatial context and non-linear relationships [13], [14].

Despite this progress, there are still gaps: (i) limited work on super-resolving coarse soil moisture data (e.g., SMAP at 9 km) to plot-scale maps using deep learning [11], [15], [16], (ii) few operational frameworks that leverage GEE for large-scale, automated data preparation and (iii) scarce studies using weak supervision, where composite satellite-based indices replace expensive field campaigns as training labels. The present work

addresses these gaps by designing a multi-modal Attention U-Net trained on a Smart Health Index that fuses vegetation, soil moisture and climate variables for the Taroudant case study.

III. METHODOLOGY

A. Study Area and Data Description

The study area is defined as a 20 km radius buffer centered on the city of Taroudant (approximately 30.470°N, 8.877°W). The area exhibits a semi-arid to arid climate, with annual precipitation typically between 200 and 400 mm and high evapotranspiration rates. Land use is dominated by irrigated citrus orchards, olive groves and rain-fed agricultural plots, all subject to increasing water stress in the Souss–Massa basin [1], [2].

The analysis focuses on the main growing season from January 1, 2023 to April 30, 2023. All data were accessed and preprocessed in Google Earth Engine.

A multi-source 16-band stack was compiled at 3 km resolution, including:

- **Sentinel-2 raw bands:** Blue, Green, Red, NIR, SWIR1, SWIR2.
- **Sentinel-2 indices:** NDVI, NDWI, SAVI.
- **Sentinel-1 radar:** VV polarization backscatter.
- **SMAP:** Root-zone soil moisture (originally 9 km).
- **Climate:** Land Surface Temperature (LST) and CHIRPS rainfall.
- **Terrain:** SRTM elevation and slope.
- **Socio-economic:** ESA WorldCover land cover and population density.

TABLE I
SUMMARY OF DATA SOURCES USED IN THE TAROUDANT CASE STUDY.

Category	Description
Optical (S2)	Six surface reflectance bands (B2, B3, B4, B8, B11, B12) at 10–20 m, resampled to 3 km.
Indices	NDVI, NDWI and SAVI derived from Sentinel-2.
Radar (S1)	VV backscatter (ascending and descending orbits, aggregated).
Soil Moisture	SMAP root-zone soil moisture, downscaled from 9 km to 3 km.
Climate	LST (temperature) and CHIRPS rainfall.
Terrain	SRTM elevation and slope.
Socio-economic	Land cover (WorldCover) and population density.

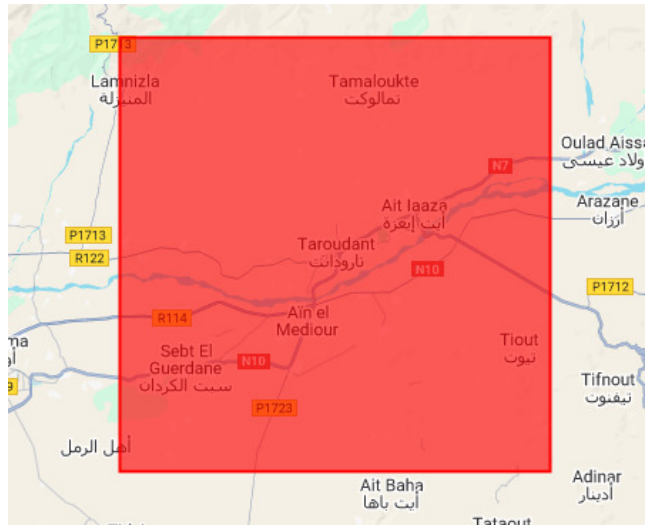

Fig. 1. Study area: 20 km buffer around Taroudant, Morocco.

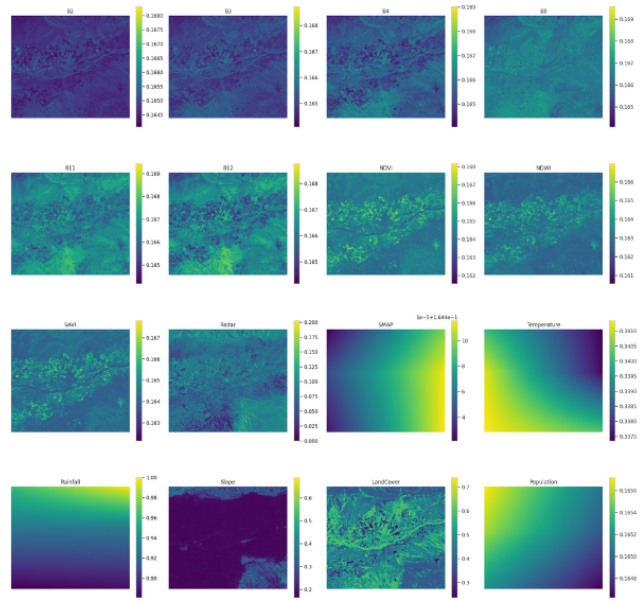

Fig. 2. Example visualization of the multi-source 16-band inputs used in the Attention U-Net.

B. Smart Health Index and Weak Supervision

Extensive field measurements of soil moisture and crop status are costly and logistically challenging. To overcome this limitation, we adopt a weakly supervised strategy by constructing a synthetic but physically consistent Smart Health Index (SHI) that serves as the training label for the deep learning model, following the idea of composite satellite-based drought indicators used in previous work [6], [7], [10].

The SHI combines vegetation greenness, root-zone moisture, rainfall and temperature into a single score:

$$\mathrm{SHI} = 0.4\times\mathrm{NDVI}+0.3\times\mathrm{SMAP}+0.2\times\mathrm{Rain}-0.1\times\mathrm{LST}. \quad (1)$$

High SHI values correspond to healthy, well-watered crops; low SHI values indicate water stress. The continuous SHI map

is then discretized into three classes: Severe Stress, Moderate Stress and Healthy. These classes are used as pixel-wise labels for training the Attention U-Net.

C. Data Pre-processing

All preprocessing steps were implemented in GEE:

- **Cloud masking:** Sentinel-2 images were filtered and masked using quality flags to remove clouds and shadows.
- **Resampling:** Coarse products such as SMAP (9 km) and CHIRPS (5 km) were upscaled to 3 km using bicubic interpolation to match Sentinel-2, in line with recent downscaling strategies [11], [15], [16].
- **Normalization:** All bands were scaled to a 0–1 range to stabilize neural network training.
- **Aggregation:** The final 10m input stack was aggregated to the target 3km resolution for the final output product, ensuring regional consistency and reduced noise for policy making.
- **Export:** The final 16-band stack and SHI labels were exported as GeoTIFF tiles for offline training.

D. Multi-Scale Resolution Considerations

The proposed framework integrates satellite observations characterized by inherently different spatial resolutions, ranging from high-resolution optical and radar imagery (10–20 m) to coarse-resolution root-zone soil moisture products (SMAP at approximately 9 km). While all inputs are resampled to a common 3 km grid for computational consistency and neural network training, this does not imply that all variables carry independent physical information at this scale.

In particular, SMAP soil moisture acts as a large-scale constraint on water availability, providing regional context rather than parcel-level measurements. The Attention U-Net leverages high-resolution predictors (Sentinel-1, Sentinel-2, terrain) to redistribute this coarse information spatially, guided by learned correlations. As a result, model outputs should be interpreted as relative water stress patterns whose effective spatial support reflects both fine-scale land surface heterogeneity and coarse-scale hydrological controls.

This multi-scale interpretation enables the production of spatially detailed maps that remain physically consistent with the resolution of subsurface moisture observations, supporting regional-to-local decision-making.

E. Attention U-Net Architecture

To better capture small, fragmented agricultural plots typical of Taroudant and to focus on stressed vegetation patterns, we employ an Attention U-Net architecture instead of a standard U-Net. The network consists of an encoder-decoder structure with skip connections; attention gates are inserted in each skip connection to suppress irrelevant background features and highlight informative regions, building upon the original U-Net [13] and Attention U-Net designs [14], [17], [18].

Fig. 3. Architecture of the Attention U-Net used for water stress mapping.

The encoder progressively downsamples the 16-band input, extracting hierarchical feature representations. The decoder upsamples and fuses these features with corresponding encoder outputs through attention-modulated skip connections, producing a three-class segmentation map at 3 km resolution.

F. Training Configuration and Evaluation Metrics

Model training was implemented in TensorFlow/Keras with a focus on stability for high-dimensional inputs and large image tiles.

1) Distributed Strategy: A multi-GPU `MirroredStrategy` was used to accelerate training. To avoid tensor shape mismatches in distributed mode, the input pipeline was configured with `drop_remainder = True`, ensuring that all batches had a fixed size of 32 by discarding incomplete final batches.

2) Hyperparameters and Callbacks: The main hyperparameters were:

- **Optimizer:** Adam, learning rate 1×10^{-4}.
- **Loss:** Categorical cross-entropy.
- **Metrics:** Accuracy and Dice coefficient to account for class imbalance.
- **Callbacks:**
  - **ModelCheckpoint:** Save best weights based on validation accuracy.
  - **ReduceLROnPlateau:** Halve the learning rate if validation loss plateaued for 5 epochs.
  - **EarlyStopping:** Stop training if no improvement was observed for 20 epochs and restore best weights.

3) Evaluation Metrics: Quantitative performance was evaluated using overall accuracy, precision, recall, F1-score and Dice coefficient, computed on a held-out test set of tiles. Confusion matrices were examined to understand class-wise errors, following common practice in segmentation and drought mapping studies [4], [5], [7].

IV. RESULTS AND DISCUSSION

A. Quantitative Model Performance

The Attention U-Net achieved high performance on the test data in mapping the three water stress classes. The main metrics are summarized in Table II.

TABLE II
ATTENTION U-NET CLASSIFICATION METRICS ON THE TEST SET.

Metric	Value	Description
Accuracy	0.9223	Correct pixel classification.
Loss	0.6923	Categorical cross-entropy loss.
Dice coefficient	0.85	Overlap between prediction and reference labels.
Precision	0.9268	Positive predictive value.
Recall	0.8862	True positive rate.
F1-score	0.9074	Harmonic mean of precision and recall.

The confusion matrix (Fig. 4) shows that most misclassifications occur between Moderate and Severe Stress classes, which is expected given their gradual transition in the landscape.

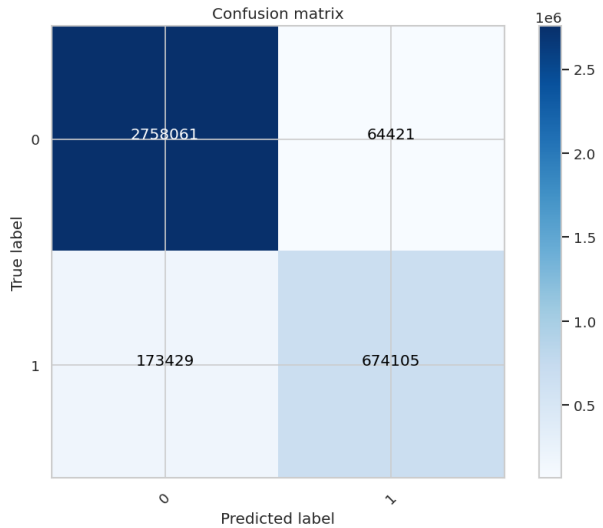

Fig. 4. Confusion matrix of Attention U-Net classification performance.

Qualitative inspection of random test tiles confirms the numerical results. The model captures fine-scale field boundaries and correctly highlights stressed patches along irrigation gradients.

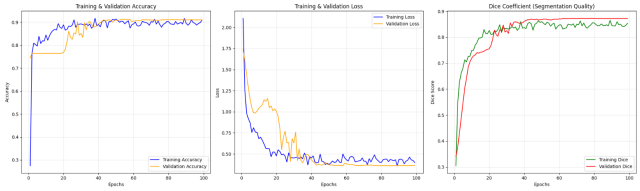

Fig. 5. Qualitative model performance: input features (left), Smart Health Index labels (center) and Attention U-Net predictions (right) for a subset of test samples.

B. Spatial Distribution of Water Stress

The resulting water stress maps reveal clear spatial patterns across the 20 km Taroudant buffer:

- **High stress areas** are concentrated in rain-fed zones and sectors with sandy, highly permeable soils, where infiltration is rapid and water retention is low.
- **Healthy zones** are primarily located in the central Souss valley, where irrigation infrastructure is dense and groundwater abstraction is highest.

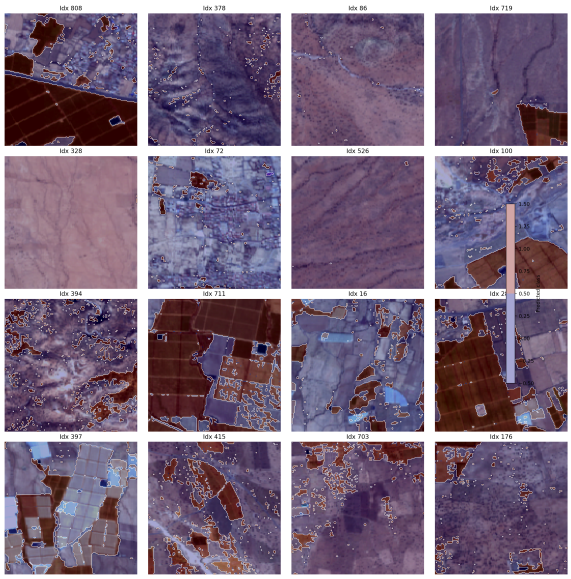

Fig. 6. Sample predictions of water stress classes over different agricultural parcels in Taroudant.

These spatial patterns align with known hydro-agricultural conditions in Taroudant and the broader Souss–Massa basin [1], [2], and confirm that the model captures meaningful environmental gradients rather than noise.

C. Interpretation of Scale and Spatial Meaning

The apparent fine spatial detail visible in the predicted stress maps reflects the contribution of high-resolution optical, radar, and terrain features. However, stress intensity should not be interpreted as an absolute parcel-scale measurement. Instead, the maps depict relative stress conditions constrained by both local surface characteristics and broader hydrological context.

This distinction is particularly important in irrigated systems such as the Souss–Massa basin, where fine-scale management practices interact with regional groundwater availability. By combining multi-scale information, the proposed framework provides a balanced representation that is sensitive to local heterogeneity while remaining consistent with basin-scale water dynamics.

D. Feature Importance and Drivers of Stress

A feature importance analysis indicates that SMAP root-zone soil moisture and NDWI are the most influential variables, followed by LST, NDVI and Sentinel-1 VV backscatter. This confirms that greenness alone (NDVI) is insufficient for drought monitoring and that combining soil moisture and plant water content yields a more robust diagnostic, in line with previous findings [4]–[7].

TABLE III
FEATURE IMPORTANCE RANKING FOR HYDRIC STRESS PREDICTION.

Feature	Importance Score
SMAP root-zone moisture	0.94
NDWI (Water Index)	0.91
LST (Land Surface Temperature)	0.88
NDVI (Vegetation Index)	0.85
Sentinel-1 radar (VV)	0.79

Beyond individual biophysical features, the model's outputs were further integrated with land cover and population density information to assess agricultural adaptation potential in a more holistic manner. This integration enables the identification of zones where water stress coincides with intensive agricultural use and higher human presence, thereby increasing vulnerability and the urgency for intervention. By jointly considering stress intensity, land-use type, and socio-economic exposure, the framework highlights areas where targeted irrigation upgrades, improved water-use efficiency, or the adoption of drought-resilient practices would yield the greatest benefits. Such a spatially explicit approach moves beyond generic drought monitoring toward localized, impact-oriented products that are better aligned with practical decision-making needs for water resource management and agricultural adaptation in semi-arid regions [6], [7], [12].

E. Role of Indices and Ancillary Variables

Table IV summarizes the main indices and variables used in the framework and their roles in the hydric stress and adaptation analysis for Taroudant.

a) Complementarity and Redundancy of Variables: Although several indices included in the framework are partially correlated, their joint use is intentional and physically motivated. Vegetation indices such as NDVI primarily capture greenness, while NDWI and SWIR bands provide complementary information on plant water content that is not directly observable through greenness alone. This distinction is critical in irrigated and perennial cropping systems, where vegetation may remain green despite emerging water stress.

Radar backscatter (Sentinel-1 VV) contributes sensitivity to surface roughness and moisture conditions independent of illumination and cloud cover, enhancing robustness under variable atmospheric conditions. Terrain variables such as slope influence runoff and infiltration processes, indirectly modulating soil moisture availability. Climate variables, including rainfall and land surface temperature, characterize atmospheric water supply and evaporative demand, respectively.

By integrating these complementary sources, the model is able to distinguish between meteorological drought, soil moisture deficits, and management-driven stress, reducing ambiguity that would arise from reliance on a single index or data source.

TABLE IV
INDICES AND THEIR ROLES IN THE HYDRIC STRESS AND ADAPTATION ANALYSIS (TAROUDANT CASE).

Index Name	Meaning / Description	Role
NDVI (Normalized Difference Vegetation Index)	Measures the difference between red and near-infrared light. Indicates greenness and vegetation density.	Primary indicator of crop health. Low values indicate senescence or severe stress.
NDWI (Normalized Difference Water Index)	Measures liquid water content inside plant leaves using NIR and SWIR bands.	Detects the plant's internal water status. A plot may be green (high NDVI) but still water-stressed (low NDWI).
SAVI (Soil Adjusted Vegetation Index)	NDVI variant with soil correction factor.	Critical in Taroudant where bright soil is visible between trees; reduces NDVI bias due to bare soil.
Sentinel-1 radar (VV backscatter)	Radar signal reflected as a function of surface roughness and moisture.	Provides information on soil and canopy structure, works in cloudy conditions and at night.
SMAP root-zone moisture	NASA product estimating deep root-zone soil moisture.	Key driver of stress: represents actual water availability for crops below the surface.
LST (Land Surface Temperature)	Heat emitted by the land surface.	High LST increases evapotranspiration and aggravates water stress.
Rainfall (CHIRPS)	Accumulated rainfall over the analysis period.	Distinguishes meteorological drought (lack of rain) from purely management-driven stress.
Slope	Terrain inclination.	Influences runoff and infiltration; steep slopes tend to retain less water.
SWIR bands (B11, B12)	Short-wave infrared sensitive to moisture and chemical composition.	Capture information on plant and soil moisture not visible in RGB.
RGB & NIR (B2, B3, B4, B8)	Natural color and near-infrared bands.	Help the network recognize objects and land-cover patterns (fields, rivers, built-up).
Land cover	Map of land-use classes.	Allows separation of agricultural land from forests, urban areas and bare soil.
Population density	Number of people per unit area.	Proxy for human vulnerability: stress in densely populated zones is more critical for planning.

F. Methodological Limitations

Several methodological limitations should be acknowledged. First, the Smart Health Index relies on fixed weights derived from expert knowledge and literature rather than site-specific calibration, which may influence class boundaries between stress levels. Second, the weakly supervised training strategy constrains the model to reproduce the SHI formulation, limiting its ability to discover alternative stress

representations.

Third, the temporal scope of the analysis is limited to a single growing season (January–April 2023), preventing assessment of interannual variability or long-term drought trends. Finally, socio-economic variables are incorporated at relatively coarse resolution, which may underestimate fine-scale vulnerability patterns.

These limitations define the intended scope of the framework as a decision-support and exploratory monitoring tool rather than a replacement for field-based hydrological measurements.

V. CONCLUSION AND PERSPECTIVES

This paper presented an Attention U-Net framework for mapping agricultural water stress and adaptation potential in the Taroudant region of Morocco. A 16-band multi-source dataset was constructed in GEE, merging Sentinel-1/2, SMAP, climate, terrain and socio-economic information. A weakly supervised Smart Health Index served as the training label, overcoming the lack of dense field measurements.

The model achieved high accuracy (91.13%) and a Dice coefficient of 0.8570 in segmenting Severe, Moderate and Healthy water stress classes at 3 km resolution. Feature importance analysis highlighted the dominant role of SMAP root-zone moisture and NDWI, emphasizing the need to go beyond greenness indices alone [4], [6], [7]. The resulting maps provide actionable evidence for irrigation scheduling and drought adaptation planning at field scale, which is particularly relevant for vulnerable basins such as Souss–Massa [1], [2].citeSeifEnnasr2016,Attar2022.

Future work will focus on (i) integrating limited in-situ measurements to refine the Smart Health Index, (ii) transferring and validating the framework in other semi-arid regions and (iii) exploring temporal models that track stress dynamics over multiple seasons.

ACKNOWLEDGMENT

The authors would like to thank the providers of Sentinel-1/2, SMAP, CHIRPS, SRTM and ESA WorldCover datasets, as well as the Google Earth Engine team for providing the cloud processing environment that made this work possible.

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